

Methods

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Selection of community forest sample

We used a stratified random approach for selecting CFs for the survey. First, we identified eight provinces for sampling, omitting Luapula and Copperbelt provinces which did not have many CFs. Our intention was to select ten CFs per province stratified by size and age (preference for older CFMAs). However, we found that three of the selected provinces had fewer than ten CFs. Hence, to maintain the number of CFs at 80, we selected 12 CFs from provinces that had more than ten. We established a cut off establishment date of 1st Jan 2024, so that all selected CFs were at least 1 yr old. In addition, we were not able to use CF size as a selection criteria, because of the low number CFs in several of the provinces. Within provinces, we selected our sample randomly from among the CFs that were at least 1 yr old.

Survey tool

We developed a survey questionnaire tool to assess the quality of CFM implementation and outcomes for SFM and sustainable development. The same tool was used in the FGDs and in the individual interviewees. The structured design enabled comparisons among the responses from different FGDs, individual interviewees and across CFs. However, interviewees were encouraged to discuss the appropriate response, and comments sections throughout the tool enabled free text responses that could capture information not listed among the options in the tool. The complete tool is provided in Appendix I. The tool was fully pre-tested at M'phansha CF in Rufunsa and feedback from the survey team, the FDG groups and the individual interviewees was incorporated into the final version of the tool.

The tool covered CF (i) governance including business activities, (ii) SFM outcomes, such as impacts on deforestation, fire frequency, and wildlife, (iii) economic transformation including the impact of CFM on access to forest resources, employment opportunities, income earning opportunities and opportunity costs, (iv) equity and inclusion, including benefit sharing arrangements and their impact on vulnerable groups, and (v) opportunities for improving CFM implementation.

In each CF forest, the team conducted four FGDs including (i) CFMG executive, and community (ii) females, (iii) males and (iv) youth. Thus, we are able to compare responses among these stakeholder groups. In addition, the survey team conducted individual interviews with key stakeholders. The aim was to interview (i) a CFMG executive member, usually the secretary or chairperson, (ii) the DFO, (iii) traditional leadership, usually a village headperson, (iv) the chairperson of a forest user-group, and (v) a representative of any external organisation supporting the CFM process, such as a conservation NGO or a carbon trading enterprise. Again, by consistently selecting interviewees from among these different types of key stakeholder, we are able to compare responses within and across CFs. However, the number of FGDs and individual interviews were able to conduct varied according to the availability of the stakeholders when we were conducting the survey (Appendix I).

Survey of community forests

The surveys were carried out according to the following schedule (Table 1).

Table 1: CF survey schedule

Province	Dates
Lusaka	13 March to 17 April 2025
Southern	8 May to 28 May 2025
Central	8 June to 21 June 2025
Eastern	8 July to 31 July 2025
Northern	10 Aug to 2 September 2025
Muchinga	10 September to 5 October
North-western	17 October to 11 November 2025
Western	26 November to 8 December 2025

During the survey, a five CFs were dropped from the sample. For various reasons the five CFs (Table 2), these CFs existed on paper, but were non-functioning on the ground.

Table 2: CFs selected but not surveyed

Province	CF	Reasons for omission
Lusaka	Chalimbana	The CF was not operational and currently exists only in the database. Only the initial establishment stages were completed.

Province	CF	Reasons for omission
Central	Kaindu	The Chief had disbanded the executive committee and hence it was not possible to organise the community
Muchinga	Nampando	The CF was not operational and currently exists only in the database. Only the initial establishment stages were completed.
Eastern	Kalumphila	This CF has not been functional since the partner (Land Alliance) withdrew in 2022. The CFMG establishment processes were not concluded.
Eastern	Koloko	This CF has not been functional since 2022 after the partner (Land Alliance) withdrew.

The final list of CFs surveyed is given in Appendix II.

The surveys were conducted by the following interviewers (Table 3). Chipso Chisonga (CIFOR-ICRAF), Edward Banda (CIFOR-ICRAF) and Thecla Masuku (Forestry Department) comprised the principle survey team with others supporting them as required. Thecla Masuku, supported by Hope Chame (Forestry Department), were responsible for conducting the community female FGDs and female individual interviewees. Given the gender norms in local communities, it was felt that female interviewees might feel inhibited to answer honestly with male interviewers.

```
cfmg |>
  mutate(
    interview_type = case_match(
      interview_type,
      "cfmg_executive" ~ "CFMG Executive",
      "cfmg_female" ~ "Community Female",
      "cfmg_male" ~ "Community Male",
      "cfmg_youth" ~ "Community Youth",
      "individual_interview" ~ "Individual interviews",
      .default = interview_type
    ),
```

```

demo_information_interviewer_name = str_to_title(str_replace_all(demo_information_interv
) |>
count(demo_information_interviewer_name, interview_type) |>
pivot_wider(names_from = interview_type, values_from = n, values_fill = 0) |>
relocate(`Individual interviews`, .after = last_col()) |>
mutate(Total = rowSums(across(where(is.numeric)))) |>
arrange(demo_information_interviewer_name) |>
gt() |>
cols_label(demo_information_interviewer_name = "Interviewer") |>
cols_align(align = "left", columns = demo_information_interviewer_name) |>
cols_align(align = "center", columns = where(is.numeric))

```

Table 3: Number of interviews conducted by each interviewer, by interview type.

Interviewer	CFMG Executive	Community Female	Community Male	Community Youth	Indiv
Chipo Chisonga	46	1	47	0	
Edward Banda	23	1	22	51	
Hockings Mambwe	3	0	0	9	
Hope Chame	1	12	0	0	
Memory Sakuwaha	0	1	0	0	
Ronald Chamwaza	0	0	0	1	
Thecla Masuku	2	61	3	8	

Analysis of the survey results

Data from the survey tool were summarised and displayed using tables, bar charts or word clouds for informative interpretation. To examine hypotheses concerning differences in responses among FGD types or interviewee roles for the individual interviews, we used Generalised Linear Mixed Models (GLMM). For binary responses (e.g. yes/no), we used binomial models and for ordered categories (e.g. decrease, no change, moderate increase, significant increase), we used Cumulative Link Mixed Models (CLMM). Reflecting the nested structure of the data with interviews nested within CFs nested within provinces, we implemented random intercepts for province and CF name nested within province (i.e. province:CF name interaction). For the FGD responses, FGD types was included in the model as a factor and we examined the pairwise differences among FGD types using Tukey Honest Significant Differences (HSDs). For the models investigating interview roles for the individual interviews, we inserted interviewee gender, interviewee age and interviewee role as factors in the models and the pairwise differences among interviewee roles were examined using Tukey HSDs.

To provide an overall summary of CFMG performance across the four thematic areas, we constructed four composite indexes. The **governance index** (0–10) counts the number of

ten governance features reported as present by the CFMG executive FGD: receipt book, bank account, financial reporting, annual reporting, issuing permits, HFOs appointed, workplan, budget, business plan, and term limits. The **SFM index** (0–5) sums five binary indicators averaged across all four FGD types: reduced forest clearing, reduced or absent charcoal production, reduced fire incidence, increased wildlife, and improved forest resource management. The **economic transformation index** (0–5) sums five binary indicators averaged across all four FGD types: improved household access to natural resources, improved access to natural resources for sale, new formal employment opportunities, collaboration with a private enterprise, and absence of opportunity costs (scored 0 if opportunity costs were reported, 1 if not). The **equity and inclusion index** (0–5) sums five binary indicators averaged across all four FGD types: equitable benefit sharing, women benefiting from CFM, women participating in decision making, transparency in benefit sharing, and existence of a grievance and redress mechanism. For each index, linear mixed models with FGD type and the governance index as predictors and random intercepts for province and CF within province were used to test whether governance and FGD type were significant predictors of performance. Mean scores for all four indexes by CFMG are provided in Appendix IV.

Remote sensing analysis

Fire dataset

To_Do

Deforestation dataset

To_Do

Analysis of remote sensed data

To_Do